RNC

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name: RENEWCOILS ENGINEERING AND SUPPLY CO . LTD
WPS No.: RNC-WPS-021 Date: 23-May-13 Revision No.: 0
Supporting PQR No.(s): RNC-PQR-021 Date: 29-Apr-13 Type(s): Manual
Welding Process(es): Gas Tungsten Arc Welding (GTAW)

JOINT (QW-402)

Joint Design: GROOVE & FILLET
Backing (Yes) ____ (No) X
Backing Material (Type): N/A

☐ Metal ☐ Nonfusing Metal ☐ Nonmetallic ☐ Other

Notes:

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. 5B Group No. 1 to P-No. 5B Group No. 1

Material Specification Type and Grade: A335 P5 , F182-F5, A234 WP5 TO A335 P5 , F182-F5, A234 WP5
Chem. Analysis and Mech. Prop.: N/A
Thickness Range :
Base Metal : Groove 1.6-14.22 Fillet ALL
Pipe Dia. Range : Groove ALL Fillet ALL
Other: N/A

FILLER METALS (QW-404)

FILLER METALS (QW-404)	GTAW	SMAW
Spec. No. (SFA)	5.28	
AWS No. (Class)	ER80S-B6	
F-No.	6	
A-No.	5	
Size of Filler Metals	Ø 2.4 ~ 3.2mm.	
Deposited Weld Metals	7.11 mm.	
Thickness Range :		
Groove	14.22 mm.	
Fillet	ALL	
Electrode-Flux (Class)	-	
Flux Trade Name	-	
Consumable Insert	-	
Other	BOHLER CM 5-IG or Equivalent	

POSITION (QW-405)

Position(s) of Groove: 6G
Welding Progression: Up X Down ____
Position(s) of Fillet: ALL

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range: 714°C~770°C ±10°C
Time Range: Holding Time 2 Hour
Notes: Heating Rate = 200°C Max/Cooling Rate = 200°C Max

PREHEAT (QW-406)

Preheat Temp. Min. 250°C
Interpass Temp. Max. 300°C
Preheat Maintenance: Continuous at 300 °C 1 HR.
Notes: Resistance heater

GAS (QW-408)

	Gas(es)	Percent Composition (Mixture)	Flow Rate
Shielding	Argon	99.99%	12-20 Ltr/Min
Trailing	-		-
Backing	Argon	99.99%	13-20 Ltr/Min

RNC WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-021 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current [ ] AC [X] DC Polarity EN(GTAW)

Amps (Range) See Below Volts (Range) See Below

Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)

(Pure Tungsten, 2% Thoriated, etc.)

Mode of Metal Transfer for GMAW N/A

(Spray arc, short circuiting arc, etc.)

Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS

Orifice or Gas Cup Size 8.0 mm - 19.0 mm

Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING

Method of Back Gouging N/A

Oscillation N/A

Contact Tube to Work Distance N/A

Multiple or Single Pass (per side) MULTIPASS

Multiple or Single Electrodes SINGLE ELECTRODE

Travel Speed (Range) See Below

Peening N/A

Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER80S-B6	2.4 mm.	DC EN	80-110	8-12	4-7	
2nd and Other	GTAW	ER80S-B6	2.4 mm.	DC EN	100-130	10-13	5-9	

We,the undersigned, certify that the statements in this record are correct and that the test welds were prepared ,welded, and tested in accordance with the requirement of **ASME-IX, 2010 latest Revision.**

Prepared by : Mr. Kittipong M. (Name) [signature] (Signed) May 23, 2013 (Date)

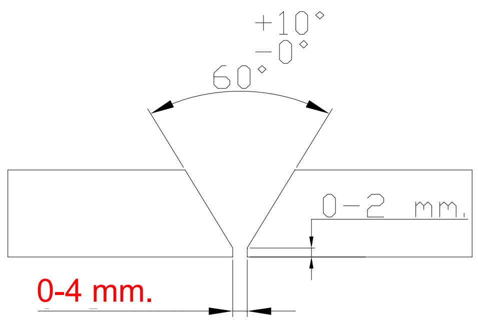
Reviewed by : ภูชิน สีหอมไชย Phuchin Seehomchai (Name) [signature] (Signed) 24/05/13. (Date)

Approved by: ภูชิน สีหอมไชย Phuchin Seehomchai (Name) [signature] (Signed) 24/05/13. (Date)

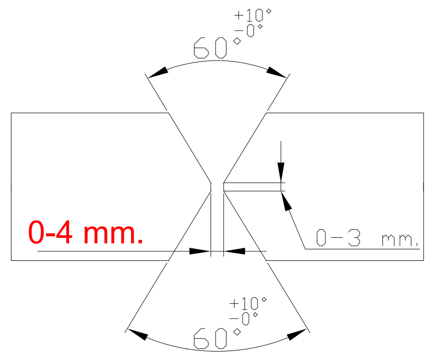
Attachment ; joint sketch.

JOINT PREPARATIONS

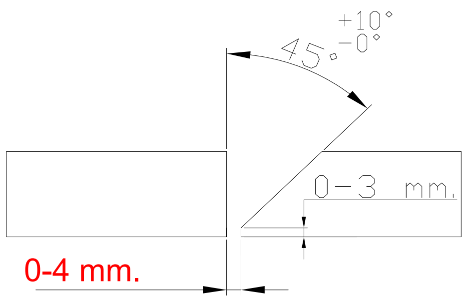
1. Single Vee Groove



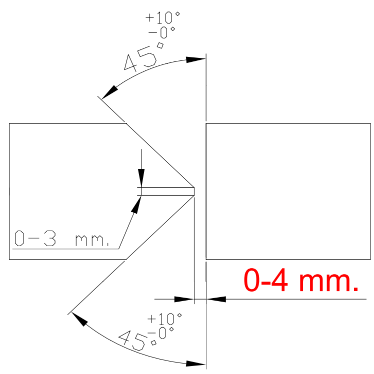
2. Double Vee Groove



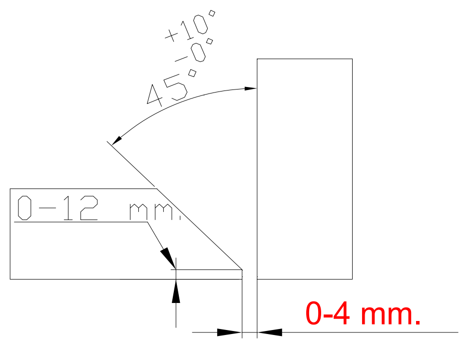
3. Single Bevel Groove



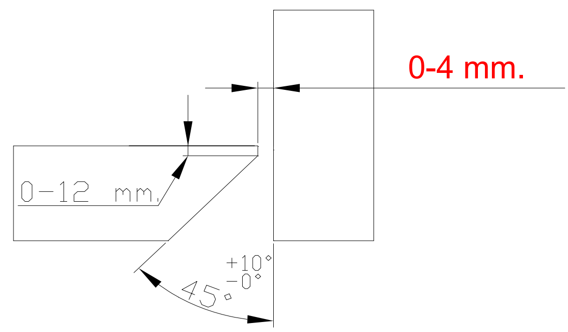
4. Double Bevel Groove



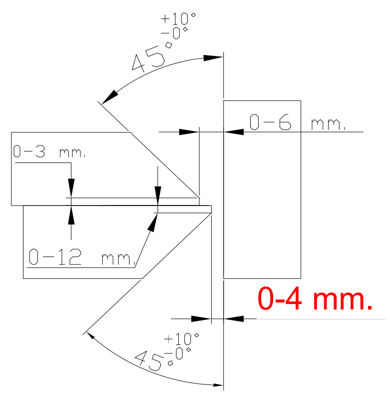
5. Bevel Groove



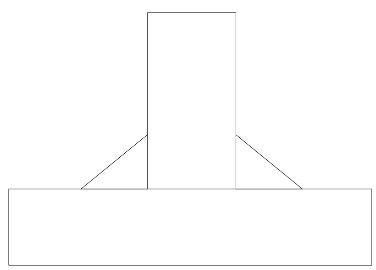
6. Bevel Groove



7. Bevel Groove



8. Fillet



PAE TECHNICAL SERVICE

Procedure Qualification Record (PQR)

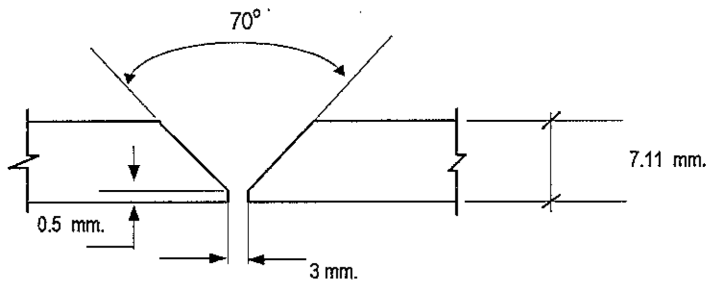
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-021

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-021	Date :	29 April 2015
WPS Number :	RNC-WPS-021		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW)		
Type (Manual, Semi-auto, or Automatic):	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)

Material Spec.	A335	To	A335
Type or Grade	P5	To	P5
P. No.	5B	To	5B
Thickness of Test Coupon	7.11 mm.		
Diameter of Test Coupon	Ø 6"		
Others			

FILLER METALS (QW-404)

	GTAW
SFA Specification	A 5.28
AWS Classification	ER80S-B6
Filler Metal F-No.	6
Weld Metal Analysis A-No.	4
Size of Filler Metal	Ø 2.4 mm.
Others	Kobelco TGS-5CM
Weld Metal Thickness	-

POSITION (QW-405)

Position of Groove	6G
Progression of Welding (Uphill or Down)	Up Hill
Others	N/A

PREHEAT (QW-406)

Preheat Temperature	220° C
Interpass Temperature (max.)	265° C
Others	N/A

POSTWELD HEAT TREATMENT (QW-407)

Temperature	740° C
Hold Time	2 HR
Others	Heating Rate 185° C/Hr
	Cooling Rate 200° C/Hr

GAS (QW-408)

	Percent Composition		
	Gas (es)	Mixture	Flow Rate
Shielding	Argon	99.99%	15 L/Min.
Trailing	-	-	-
Backing	Argon	99.99%	17 L/Min.

ELECTRICAL CHARACTERISTICS (QW-409)

Current	DC (GTAW)		
Polarity	EN		
Amperage	80 - 130	Voltage	9.5 - 11
Tungsten Electrode Size	ø 2.4 mm. 2% thoriated.EWTh-2		
Others	N/A		

TECHNIQUE (QW-410)

Travel Speed	5.5 - 6.5 cm/min.
String or Weave Bead	Both
Oscillation	N/A
Multiple or single Pass	Multiple Pass
Multiple or Single Electrodes	Single Electrodes
Others	N/A

PAE Technical Service Company Limited

Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number-: RNC-PQR-021

Tensile Test Report No.TS-13-05-049

Specimen	Width (mm)	Thickness (mm)	Area (mm^2)	Ultimate Total Load (KN)	Ultimate Strength (N/mm^2)	Type of Failure & Loc.
1263/13 TR1	19.37	6.82	132.103	64.214	486.090	OUT OF WELD
1263/13 TR2	19.36	6.79	131.454	64.567	491.176	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-05-008

Type and Figure Number	Type of Bend	Indication	Results
1263/13 BF1	Transverse face bend	No Open Discontinuity	Accepted
1263/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
1263/13 BR1	Transverse root bend	No Open Discontinuity	Accepted
1263/13 BR2	Transverse root bend	Open Discontinuity 1.0 mm	Accepted

Toughness Tests (QW-170) Report No.IM-13-05-016

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Average (Joules)	Lateral Expansion Mils
1263/13-1	Weld Metal	V-Notch	- 20 °C	117.9	120.53	
1263/13-2	Weld Metal	V-Notch	- 20 °C	119.69		
1263/13-3	Weld Metal	V-Notch	- 20 °C	124.01		
1263/13-1	HAZ	V-Notch	- 20 °C	186.39	181.19	
1263/13-2	HAZ	V-Notch	- 20 °C	189.53		
1263/13-3	HAZ	V-Notch	- 20 °C	167.64		
1263/13-1	Base Metal	V-Notch	- 20 °C	142.46	108.93	
1263/13-2	Base Metal	V-Notch	- 20 °C	86.96		
1263/13-3	Base Metal	V-Notch	- 20 °C	97.36		

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A

Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR-010

Hardness Testing Report No. HV-13-05-003

Macrostructure Report No. MA-13-05-001

Welders Name Mr.Amnaj N. Welder Number W-061

Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :

Inspected by : Mr.Wanlop N.

Certified by : Mr.Chaloemkiet R.

PAE TECHNICAL SERVICE

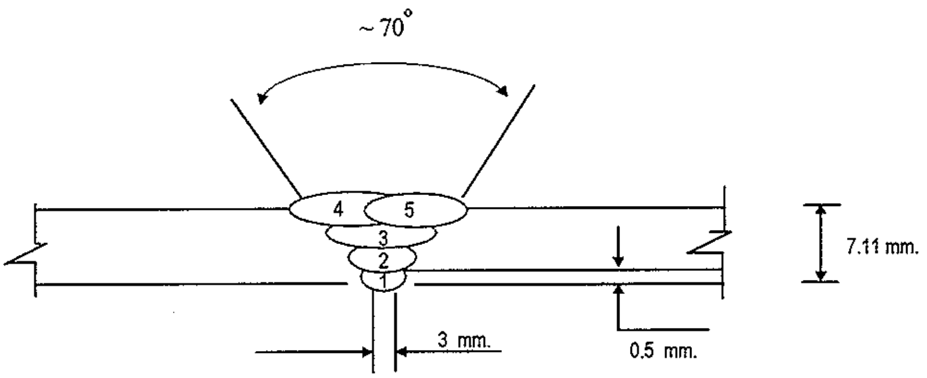
Date : 20 May 2013

PAE Technical Service Company Limited Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-021
		WPS No. : RNC-WPS-021
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST
		CONDITION AND RESULT

Welding Sequences for RNC-PQR-021 (Butt Joint)



Pipe Dia. 6"T.7.11

Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave Bead	Wire Feed Speed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
1	GTAW	ER80S-B6	2.4	80	9.5	DCEN	-	-	5.50	8.29	15	17
2	GTAW	ER80S-B6	2.4	130	9.5	DCEN	-	-	6.20	11.95	15	17
3	GTAW	ER80S-B6	2.4	130	10.2	DCEN	-	-	6.30	12.60	15	17
4	GTAW	ER80S-B6	2.4	130	11	DCEN	-	-	6.00	14.00	15	17
5	GTAW	ER80S-B6	2.4	130	11	DCEN	-	-	6.90	13.20	15	17

PAE Technical Service Company Limited

Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

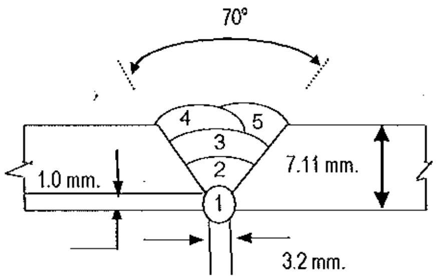
PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINERING CONSULTANT
69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 682208-9

WELDER PROCEDURE QUALIFICATION RECORD

Client :	Project Name : MINI SHUTDOWN PTTGC	Location : RNC Work Shop	Report No. PQR-008
Renew Coils Engineering & Supply Co.,Ltd.	Customer Order No. : N/A	WPS No. : RNC-WPS-021	Date : 29-Apr-13

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material Spec	Material Size	Electrode Specification	Results Visual	Results RT
W-061	MR.ANANJ N.	GTAW	6G	ASME IX	A335 P5	ø 6" Thk. 7.11 mm.	ER80S-B6	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal Class	Filler Metal Dia. (mm.)	Current Type Polarity	Current Amp	Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding GAS	Shielding Flow Rate (l/min)	Backing GAS	Backing Flow Rate (l/min)
1	GTAW	ER80S-B6	3.2	DCEN	80	9.5	5.5	205	8.3	Argon	15	Argon	17
2	GTAW	ER80S-B6	3.2	DCEN	130	9.5	6.2	207	12.0	Argon	15	Argon	17
3	GTAW	ER80S-B6	3.2	DCEP	130	10.2	6.3	212	12.6	Argon	15	Argon	17
4	GTAW	ER80S-B6	3.2	DCEP	130	11	6.0	215	14.0	Argon	15	Argon	17
5	GTAW	ER80S-B6	3.2	DCEP	130	11	6.9	215	13.2	Argon	15	Argon	17

PAE Inspection By	Witness By	Approved By
SIGNATURE :	SIGNATURE :	SIGNATURE :
NAME : Mr.Sayam S.	NAME : Mr. Chatchawan S.	NAME : ภูชิน สีหอมไชย / Phuchin Seehomchai
DATE : 30-Apr-2013	DATE : 3/05/13.	DATE :

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

LPN PLATE MILL

0451

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1

Report No. TS-13-05-049

Revision No. 0

Test Product	PQR No. RNC-PQR-021	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P5	Type of Test	Reduced Section Tension
Dimension	Ø 6" #40 (7.11 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	6 May 2013	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	8 May 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm^2)	Gauge Length (mm)	Load (kN)	Strength (N/mm^2)	Load (kN)	Strength (N/mm^2)		
1	1263/13 TR1	N/A	6.82	19.37	132.103	N/A	N/A	N/A	64.214	486.090	N/A	OUT OF WELD
2	1263/13 TR2	N/A	6.79	19.36	131.454	N/A	N/A	N/A	64.567	491.176	N/A	OUT OF WELD

The unit of N/mm^2 is equivalent to MPa

Additional details : WPS No. RNC-WPS-018

Welding Process : GTAW

Filler/Electrode Spec't : ER80S-B6

Reported by : N. [signature]

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 20 / 05 / 13

Approved by : N. [signature]

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 20 / 05 / 13

LPN PLATE MILL

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

30454

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-05-008

Revision No. 0

Test Product	PQR No. RNC-PQR-021	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P5	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 6" #40 (7.11 mm Thk.)	Diameter of Meudrel	28 mm
Date Received	6 May 2013	Bend Angle	180 Degree.
Date Tested	8 May 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	1263/13 BF1	Transverse face bend	7.25x37.95x162.18 mm	No Open Discontinuity	N/A	-
2	1263/13 BF2	Transverse face bend	7.25x38.03x163.48 mm	No Open Discontinuity	N/A	-
3	1263/13 BR1	Transverse root bend	7.20x38.06x163.47 mm	No Open Discontinuity	N/A	-
4	1263/13 BR2	Transverse root bend	7.21x37.92x162.78 mm	Open Discontinuity	WELD ZONE	1.0 mm

Additional details : WPS No. RNC-WPS-018

Welding Process : GTAW

Filler/Electrode Spec't : ER80S-B6

Reported by : N. [signature]

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 20 / 05 / 13

Approved by : N. [signature]

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 20 / 05 / 13

LPN PLATE MILL

LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

0457

TESTING No. 0134

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-05-016

Revision No. 0

Test Product	PQR No. RNC-PQR-021	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a[ε1]
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P5	Type of Test	Charpy V-Notch Impact
Dimension	Ø 6" #40 (7.11 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	6 May 2013	Equipment 's Capacity	542 Joules
Date Tested	8 May 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension Thickness (mm)	Width (mm)	Length (mm)	Results Energy Absorbed (Joules)	Average (Joules)	Location of Test	Remark
1	1263/13-1	-20°C	5.008	10.009	54.84	186.39	181.19	HAZ	-
2	1263/13-2	-20°C	5.010	10.012	54.94	189.53			
3	1263/13-3	-20°C	5.011	10.013	54.89	167.64			
4	1263/13-1	-20°C	5.009	10.017	54.87	142.46	108.93	Base Metal	-
5	1263/13-2	-20°C	5.010	10.017	54.88	86.96			
6	1263/13-3	-20°C	5.009	10.013	54.89	97.36			
7	1263/13-1	-20°C	5.013	10.010	54.91	117.90	120.53	Weld Metal	-
8	1263/13-2	-20°C	5.014	10.014	54.96	119.69			
9	1263/13-3	-20°C	5.011	10.011	54.97	124.01			

Additional details : WPS No. RNC-WPS-018

Welding Process : GTAW

Filler/Electrode Spec't : ER80S-B6

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date ..20../..05../..13...

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date ..20../..05../..13...

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

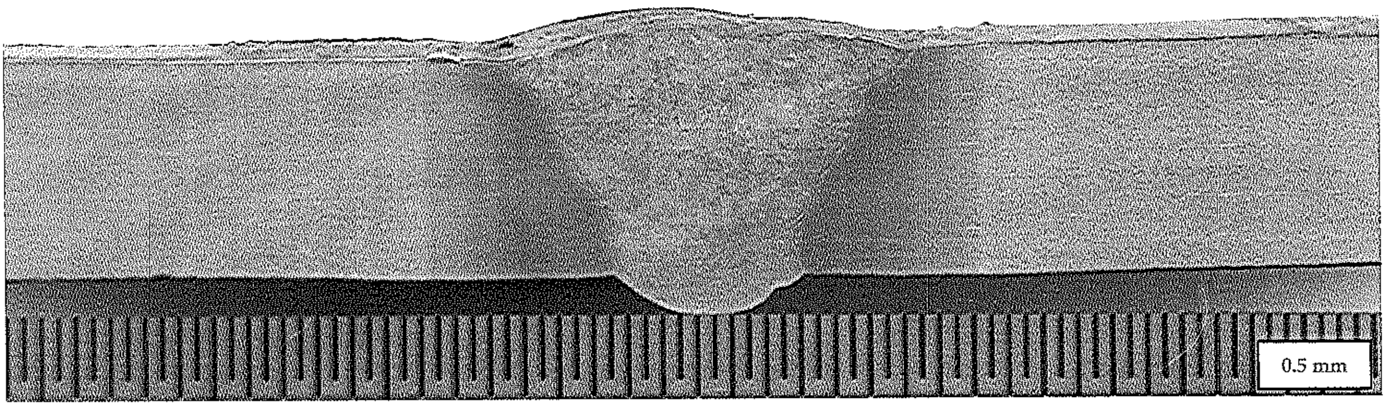
Macrostructure Analysis Report

Report No. MA-13-05-001
Revision No.0

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Test Product	PQR No.RNC-PQR-021
Test No.	1263/13 MA
Test Location	-
Material Specification	A335P5
Dimension	Ø6"#40 (7.11 mm Thk.)
Welding Process/Position	GTAW
Type of Joint	Butt Joint
Welder Name / No.	N/A
Date Received	6 May 2013

Material Characterization	Pipe to Pipe
Equipment used	Microscope upto 40X (times)
Serial No.	LEICA 0650854912
Method of Preparation	ASME Secction IX : 2010 edition
Test Method	ASME Secction IX : 2010 edition
Standard of Test	ASME Secction IX : 2010 edition
Etchant Reagent	25 ml HNO_3 + 75 ml H_2O
Time of Etching	5 minutes
Date Tested	6 May 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : -

Reported by : ....
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 12../..05./..13

Approved by : ....
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 12../..05./..13

LPN PLATE MILL

Notes

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.
4. This report is tested for LPN Metallurgical Research Center (Thailand) Co.,Ltd.

LPN
LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

13329

Hardness Testing Report

Report No. HV-13-05-003

Revision No. 0

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

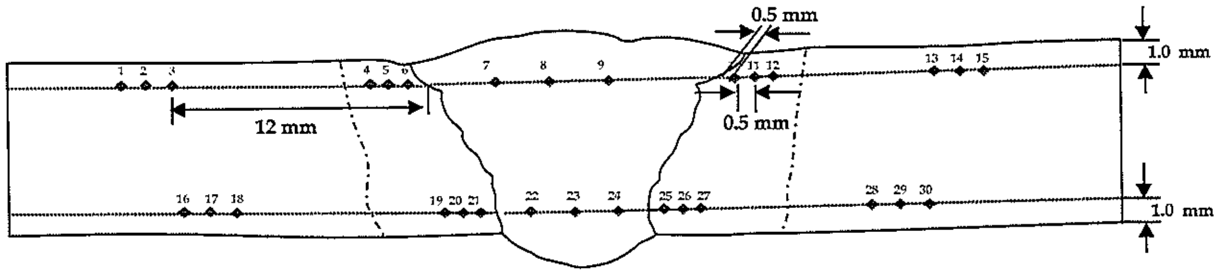
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Test Product	PQR No.RNC-PQR-021
Test No.	1263/13 HN
Project Name	N/A
Project No.	N/A
Material Specification	A335P5
Dimension	Ø6"#40 (7.11 mm Thk.)
Welding Process/Position	GTAW
Type of Joint	Butt Joint
Date Received	6 May 2013
Date Tested	6 May 2013

Material Characterization	Pipe to Pipe
Test Method	ASTM E92-82 (Reapproved 2003)$^{\varepsilon 2}$
Standard of	ASME section IX: 2010 edition
Type of test	Vicker Hardness Test
Equipment	Wilson Wolpert , 432 SVD (V32D219)
Indentor Size	Pyramid diamond, angle of 136°
Test Load	HV 10 Kgf.
Total Force Dwell Time	15 s
Cal. Block	No. A60023 No. A62072
Ambient Temperature	27.3 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)
1	BASE	133.1	16	BASE	136.1	******************************		
2	BASE	131.2	17	BASE	135.8			
3	BASE	133.7	18	BASE	136.5			
4	HAZ	187.7	19	HAZ	196.4			
5	HAZ	203.9	20	HAZ	209.0			
6	HAZ	206.2	21	HAZ	214.4			
7	WELD	196.0	22	WELD	194.7			
8	WELD	195.2	23	WELD	193.9			
9	WELD	200.7	24	WELD	204.5			
10	HAZ	193.8	25	HAZ	209.2			
11	HAZ	191.4	26	HAZ	210.9			
12	HAZ	137.2	27	HAZ	206.4			
13	BASE	130.6	28	BASE	131.6			
14	BASE	132.2	29	BASE	131.8			
15	BASE	130.8	30	BASE	131.8			

Additional details : -

Reported by : N. [signature]

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date : 20/05/13

Approved by : [signature]

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 20/05/13

LPN PLATE MILL

Notes

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PAE TECHNICAL SERVICE

PAE TECHNICAL SERVICE COMPANY LIMITED

NDT, Testing, Inspection & Engineering Consultant

69 Soi On-nuch 64 (Suksamarn), Srinakarin Road, Suanluang, Bangkok. 10250 THAILAND

Tel : BKK : 02)322-0222, Rayong : 038)682208-9, Sriracha : 038) 768923, 768924

POSITIVE MATERIAL IDENTIFICATION REPORT

Client : Renew Coils Engineering & Supply Co.,Ltd.	**Analyzer Model :** NITON XL3t 800	**Test Date :** 16-May-13
Project : Mini Shut Down PTTGC	**Analyzer Serial No.:** 39210	**Report No.** PMI-001
Procedure No. : PAE-PMI-007 Rev.0	**Location :** RNC SHOP	**Attached Report :** -
WPS No. RNC-WPS-021		

Item	Dwg No. / Line No. / Part	Location	Pipe Dia (Inc)	Thk.	Welder No.	Alloy ID.	Chemical Compositions as Found (%)					Judgment	
							Cr	Mn	Fe	Mo	Ni	Yes	No
1.1	RNC-PQR-021	Base 1	6"	7.11	-	P 5	4.56	0.47	94.25	0.48	0.00	✓	
1.2	RNC-PQR-021	Weld	6"	7.11	-	P 5	5.38	0.58	93.36	0.59	0.00	✓	
1.3	RNC-PQR-021	Base 2	6"	7.11	-	P 5	4.51	0.50	94.45	0.49	0.00	✓	

Total : 3 Points

NOTE :
- The Composition material Analysis was carried out on Surface of material
- The result of inspection accordance with ASTM table1 specification.

Mat'l Spec. A335 P5	**Chemical Range Base 1 :**	Cr. 4.00 - 6.00	Mo. 0.45 - 0.65	Mn. 0.30 - 0.60
	Chemical Range Weld :	Cr. 4.00 - 6.00	Mo. 0.45 - 0.65	Mn. 0.30 - 0.60
	Chemical Range Base 2 :	Cr. 4.00 - 6.00	Mo. 0.45 - 0.65	Mn. 0.30 - 0.60

	PAE INSPECTED BY	REVIEWED / WITNESSED BY	APPROVAL BY
SIGNATURE :			
NAME :	Mr. Wanlop N.	Kittipong M.	ภูชิน สีหอมไชย Phuchin Seehomchai
DATE :	16-May-13	23-May-13	

PAE TECHNICAL SERVICE CO.,LTD.
NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT
TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)682208-9, Fax : 038)682206, SRIRACHA : 038)330220, 330229

RADIOGRAPHIC EXAMINATION REPORT

INSPECTION & TEST RECORDS

REPORT No.	PQR-010		
DWG.No.	-		
DATE :	29-Apr-13		
CLIENT :	Renew Coils Engineering & Supply Co.,Ltd.	PROJECT NAME :	MINI SHUTDOWN PTTGC

PROCEDURE No. :	PAE-RT-001 (Rev.0)	EXPOSURE TIME	1 min. 24 sec.	SOURCE TO FILM DISTANCE	6" Inch. / - mm.
RADIATION SOURCE	X-ray (-) MAX - kVG. / γ - ray (Ir-192)	DENSITY RANGE	2.0 - 3.5	TYPE OF PENETRAMETER	ASTM-1B
		SENSITIVITY	2%		
SOURCE RANGE	20.51 Ci.	FILM TYPE	KODAK "MX"	FILM PROCESSING AUTO	-
SOURCE SIZE	3 x 3 mm.	FILM SIZE	3.5" x 8.5"	FILM PROCESSING MANUAL	✓
TUBE VOLTAGE	- kVP.	INTENSIFYING SCREEN FRONT	0.005 Inch.	MATERIAL	A335 P5
TUBE CURRENT	- mA	INTENSIFYING SCREEN BACK	0.005 Inch.	THICKNESS	7.11 mm.

APPLICABLE STANDARD : ASME Section V Article 2 — ACCEPTANCE CRITERIA : ASME-IX

RADIOGRAPHIC TECHNIQUE : ☐ SWSI ☑ DWSI ☐ DWDI

NAME / WQT	Position No.	Pipe/Plate Size Dia	Pipe/Plate Size Thk(mm)	Welder No.	Film No.	Interpretation	Results	Remarks
GTAW / WPS No.RNC-WPS-021								
MR.AMNAJ N.	6G	6"	7.11	W-061	A-B	NAD	ACC	Brfore PWHT
					B-C	NAD	ACC	
					C-D	NAD	ACC	
					D-A	NAD	ACC	

Total 3.5"x8.5" 4 Films
3.5"x17" - Films

ABBREVIATIONS :

ACC : Accepted	WH : Worm Hole	LP : Lack of Penetration	
REJ : Rejected	C : Crack	LF : Lack Of Fusion	CRP : Concave Root Penetration
CP : Cluster Porosity	P : Porosity	IP : Incomplete Penetration	ERP : Excess Root Penetration
SF : Surface Flaw	UC : Undercutting	TI : Tungsten Inclusion	NAD : No Apparent Defects
EUC : External Udercut	SL : Slag Inclusion	BT : Burn Through	RS : Reshoot

	PAE INSPECTED BY	REVIEWED BY	APPROVED BY
SIGNATURE :			
NAME :	Mr.Sarawut P.	Mr. Chatchawan S	ภูชิน สีหอมไชย Phuchin Seehomchai
DATE :	30-Apr-13	30/04/13	

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)682208-9, Fax : 038)682206, SRIRACHA : 038)330220, 330229

REPORT No.	WQT-041	RADIOGRAPHIC EXAMINATION REPORT	INSPECTION & TEST RECORDS
DWG.No.	-		
DATE :	30-Apr-13		

CLIENT :	Renew Coils Engineering & Supply Co.,Ltd.	PROJECT NAME :	Welder Supply

PROCEDURE No. :	PAE-RT-001 (Rev.0)	EXPOSURE TIME	- min. 36 sec.	DISTANCE	6" Inch. / - mm.
RADIATION SOURCE	X-ray (-) MAX - kVG. γ - ray (Ir-192)	DENSITY RANGE	2.0 - 3.5	TYPE OF PENETRAMETER	ASTM-1B
		SENSITIVITY	2%		
SOURCE RANGE	40.99 Ci.	FILM TYPE	KODAK " MX "	FILM PROCESSING AUTO	-
SOURCE SIZE	3 x 3 mm.	FILM SIZE	3.5" x 8.5"	FILM PROCESSING MANUAL	✓
TUBE VOLTAGE	- kVP.	INTENSIFYING SCREEN FRONT	0.005 Inch.	MATERIAL	CS
TUBE CURRENT	- mA	INTENSIFYING SCREEN BACK	0.005 Inch.	THICKNESS	7.11 mm.
APPLICABLE STANDARD :	ASME Section V Article 2			ACCEPTANCE CRITERIA :	ASME-IX

RADIOGRAPHIC TECHNIQUE : ☐ SWSI ☑ DWSI ☐ DWDI

NAME / WQT	Position No.	Pipe/Plate Size Dia	Pipe/Plate Size Thk(mm).	Welder No.	Film No.	Interpretation	Results	Remarks
WQT / PQR+GTAW / WPS No.RNC-021								
MR.AMNAJ N.	6G	6"	7.11	-	A-B	NAD	ACC	After PWHT
					B-C	NAD	ACC	
					C-D	NAD	ACC	
					D-A	NAD	ACC	

Total 3.5"x8.5" 4 Films

3.5"x17" - Films

ABBREVIATIONS :

ACC : Accepted	C : Crack	LF : Lack Of Fusion	CRP : Concave Root Penetration
REJ : Rejected	P : Porosity	IP : Incomplete Penetration	ERP : Excess Root Penetration
SF : Surface Flaw	UC : Undercutting	TI : Tungsten Inclusion	NAD : No Apparent Defects
	SL : Slag Inclusion	BT : Burn Through	RS : Reshoot

PAE INSPECTED BY	REVIEWED BY	APPROVED BY
SIGNATURE :	SIGNATURE :	SIGNATURE :
NAME : Mr.Sarawut P.	NAME : Mr. Somsak	NAME : Jiyas h.
DATE : 1-May-13	DATE : 05/05/56	DATE : 4/5/13

BOOK NO. 030   PAGE NO. 1486

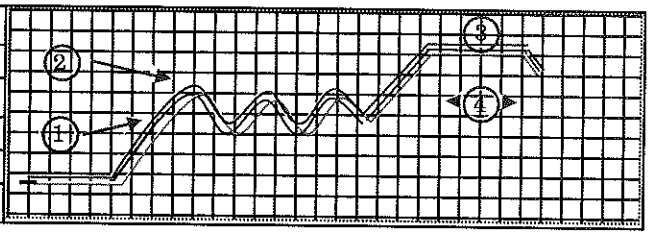
GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.

PREHEAT REPORT

CLIENT:	RENEW COILS ENGINEERING AND SUPPLY CO.-LTD.
PROJECT:	MINI SHUT DOWN PTTGC 2013
CONTRACTOR:	GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.
PREHEAT METHOD:	○ GAS ⊘ ELECTRIC COMPLETION DATE: 29-04-2013
LOCATION:	⊘ SHOP ○ FIELD REPORT NO. PH-003
RECORDER NO:	EH-07ZA194 PAGE NO. 1
CHART SPEED:	⊘ 25 MM/Hr ○ 50 MM/Hr ○ 100 MM/Hr

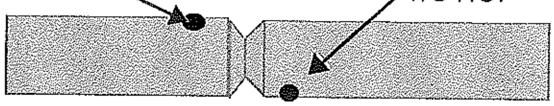
RESULT RECORDER CHART TO BE ATTACHED

DESCRIPTION	REQUEST	ACTUAL
1) PREHEAT TEMP (C° MIN)	200-250	220
2) INTERPASS TEMP (C° MAX)	300	265
3) POSTHEAT TEMP (C°)	300	300
4) POSTHEAT TIME (Hr)	1 HR.	1 HR.
MATERIAL: A335 P5		



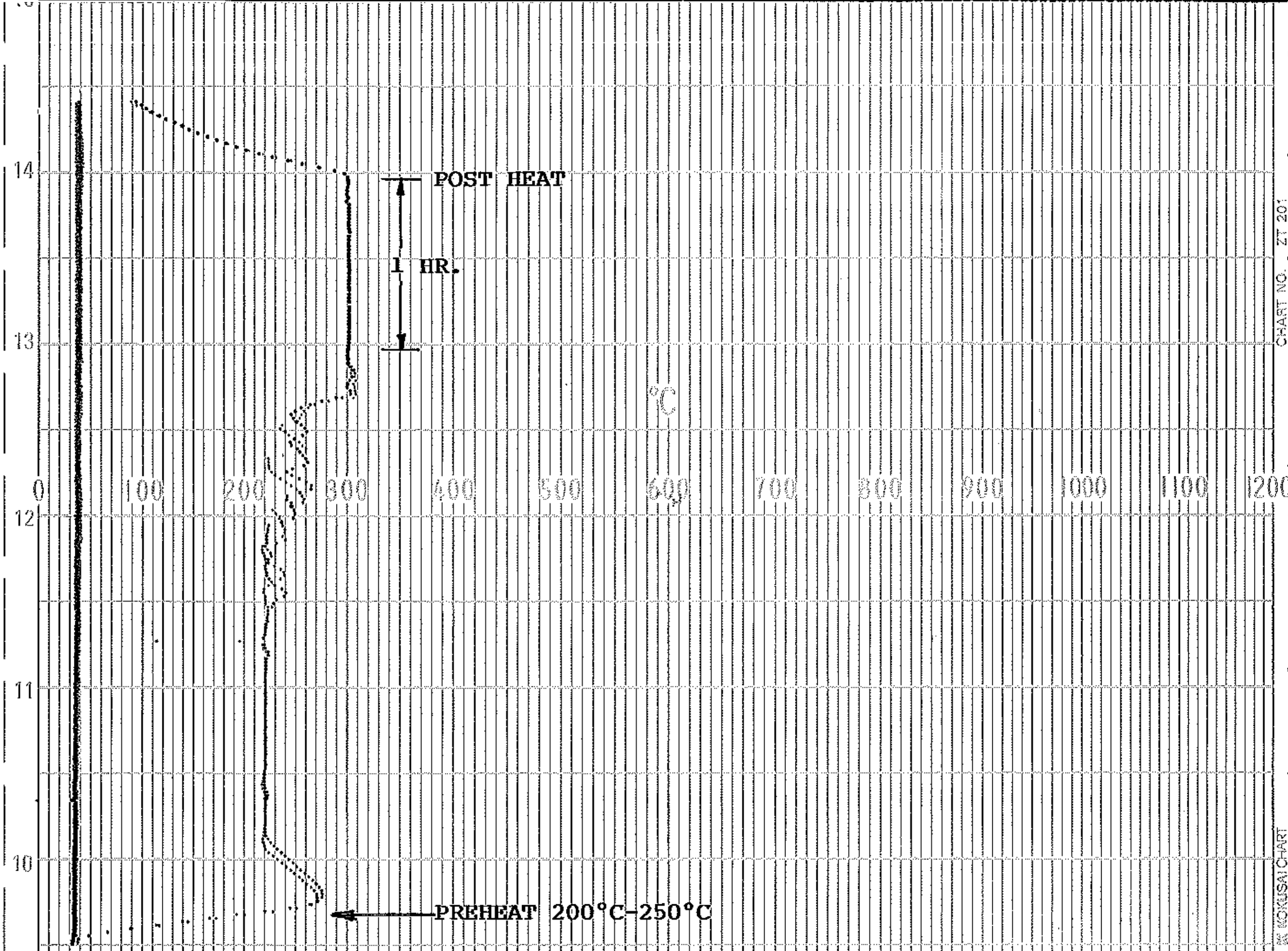
LINE NO.& CLASS & SHT NO.& SPL.NO. DWG.NO & TANK NO.	WELD NO.	SIZE	THK. (MM)	WELD TYPE	T/C NO.	REMARK
RNC-WPS-021		6"	7.11	BW	1,2	

T/C NO.   T/C NO.



TOTAL	1	JOINTS.
TOTAL	6	DB.

CHECKED BY GLOBAL HEAT TREATMENT (THAILAND)		APPROVED BY		APPROVED BY	
SIGN:		SIGN:		SIGN:	
NAME:	KOMKHUN D.	NAME:	Mr. Chatchawan S.	NAME:	ภูชิน สีหอมไชย Phuchin Seehomchai
DATE:	29-04-2013	DATE:	30/04/13	DATE:	



DATE. 29-04-2013

CHART NO. PH - 003

RECORDER NO. EH 07ZA194

CHART SPEED.25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

PROJECT. MINI SHUT DOWN PTTGC 2013

LINE NO. RNC-WPS-021

SIZE. 6" THK. 7.11 mm.

MATERIAL. A335 P5

START TIME. 11.30 AM.

TECHNICIAN. MR. CHALERMCHAI NG.

TC	COLOUR
1.	RED
2.	BLACK

BOOK NO. 011 | NUMBER 17

GLOBAL HEAT TREATMENT[THAILAND] CO.,LTD.

HEAT TREATMENT REPORT

Client :	RENEW COILS ENGINEERING & SUPPLY CO.,LTD.			Page :	1
Contractor :	GLOBAL HEAT TREATMENT[THAILAND] CO.,LTD			Report No. :	PW-003
Project :	MINI SHUT DOWN PTTGC 2013			Project No. :	
Description :	☑-PWHT	☐-Normalize	☐-Annealing	☐-Stabilize	☐-Solution ☐-Dry-out
Location :	☑-Shop		☐-Field	Recorder No.:	EH-07ZA194
Method by:	☑-Electric	☐-Gas Burner	☐-Oil burner	Material:	A335 P5
Method Type:	☑-Local	☐-Furnace	☐-Internal Firing		Unit:
Chart Speed :	☑-25 mm/Hr.		☐-50 mm/Hr.		☐-100 mm/Hr.
Completion Date:	30-04-2013		Procedure No.		

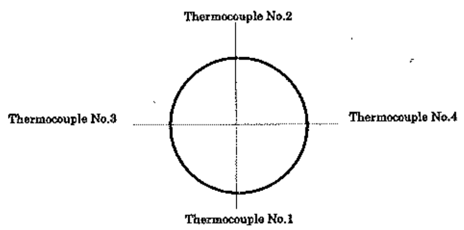
HEAT TREATMENT CONDITION/RESULTS

Description	Loading Temp[1] (°C)	Heating Rate[2] (°C/Hr.)	Holding Temp[3] (Min.-Max.)	Holding Time[4] (Hr.)	Cooling Rate[5] (°C/Hr.)	Unloading Temp[6] (°C)
Request	300	200	710-770	2 HR.	200	300
Actual	300	185	740	2 HR.	200	300
Request						
Actual						
Request						
Actual						

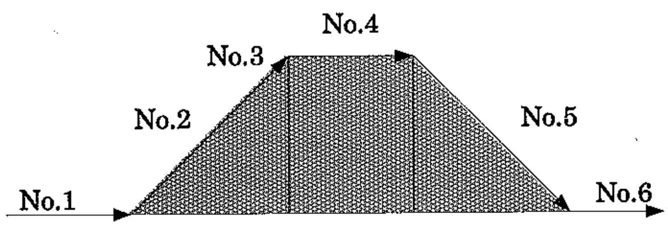
DESCRIPTION

Line No./Dwg.No./Class No./Sht.no./Spool No.	Weld No.	Size(")	THK.(MM)	Weld Type	T/C No.	Remark	
RNC-WPS-021		6"	7.11	BW	1,2		
	Total:	1		Joint.	Total:	6	DB.
	Total:			EA.	Total:		Kg.

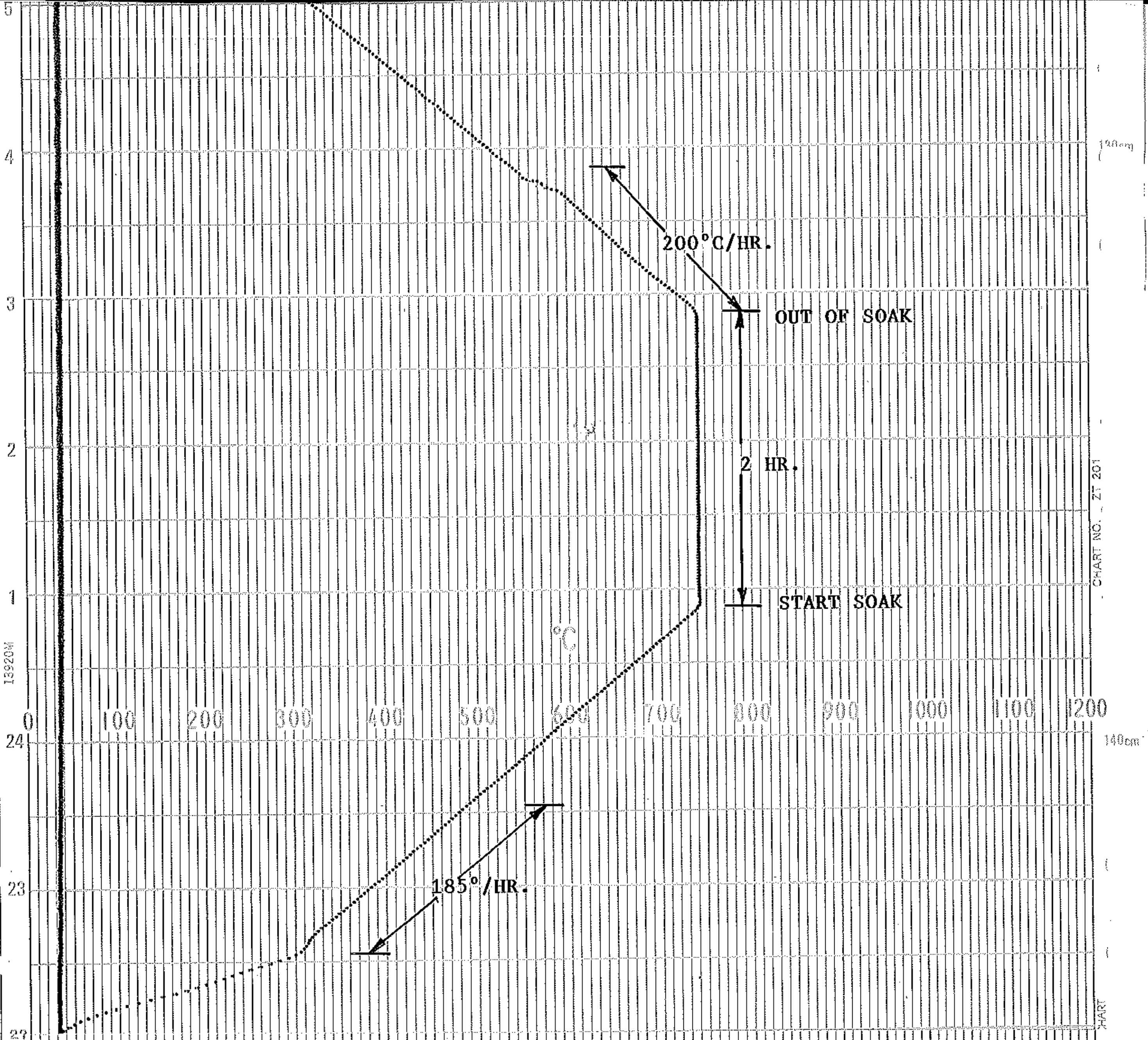
THERMOCOUPLE LOCATION



HEAT TREATMENT PROFILE



	Global Heat Treatment[Thailand] Co., Ltd. Checked By	Approve By	Approved by	Approved by
Signature:				
Name:	KOMKHUN D.	Mr. Chatchawan	ภูชิน สีหอมไชย Phuchin Seehomchai	ภูชิน สีหอมไชย Phuchin Seehomchai
Date:	30-04-2013	4/05/13		



DATE. 30-04-2013

TC	COLOUR
1.	RED
2.	BLACK

CHART NO. PW - 003

RECORDER NO. EH 07ZA194

CHART SPEED.25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO. LTD.

PROJECT. MINI SHUT DOWN PTTGC 2013

LINE NO. RNC-WPS-021

SIZE 6" THK. 7.11 mm.

MATERIAL. A335 P5

START TIME. 09.20 AM.

TECHNICIAN. MR. PHANOMTUAN CH.

Böhler Welding GROUP

Abnahmeprüfzeugnis 3.1

Inspection certificate 3.1
nach / as per: EN 10204
Nr. / No.: 2-2012-25-2198182
Rev. 0 Seite / page: 1 von/of 1

Bestell-Nr.	Order No.	email	vom / of	15.12.2011 100049
Auftrags-Nr.	Works order	1002118367		
Lieferschein/Pos./Split	Dispatch note/pos./Split	2002203341 / 0020 / 900003	vom / of	15.12.2011
Prüfgegenstand	Test object	Schweißstab welding rod		10521
Handelsname	Trade designation	**BOEHLER CM 5-IG**		X373EW02
Produktkennzeichnung	Marking of product			
Normbezeichnung	Standard classification	EN ISO 21952-A - W CrMo5 Si / AWS A5.28-05: ER80S-B6		
Abmessung	Dimension	2.40 x 1000 mm (.094" x 39")		
Charge	Heat No.	145351		
Liefermenge	Quantity delivered	405 KG (893 LB)		
Anforderungen	Requirements			

Chemische Analyse in % / Chemical composition in % — Stab / rod

Charge Heat No.	C	Si	Mn	P	S	Cr	Mo	Ni	V	Cu	Al					
145351	0.070	0.45	0.52	0.010	0.005	5.67	0.56	0.08	0.010	0.07	0.005					

Zugversuch / Tensile Test nach / according to ASTM E8 — Probenvorbereitung / Specimen preparation AWS B4.0

	Prüftemp. T Test temp. °C	Streckgrenze Reh Yield point MPa	Dehngrenze Rp Yield strength 0.2% MPa	1.0%	Zugfestigkeit Rm Tensile strength MPa	Dehnung A Elongation % Lo=4d	Einschnr. Z Reduction %	Bemerkung Remarks
Minimum	20		470		550	17		730-760°C/1h
Maximum								

Kerbschlagv. / Impact Test nach / according to ASTM E23 — Probenvorbereitung / Specimen preparation AWS B4.0

	Prüftemp. T Test temp. °C	Kerbschlagarbeit Mindest. KV Absorbed energy minimum values J	Mittelwert Average J	Lat. Breitg. Lateral expansion mm	Krist. Bruchanteil Shear fracture %	Bemerkung Remarks
	20°C (68°F)	47				730-760°C/1h

Bemerkung / Remarks: The product BOEHLER CM 5-IG meets the requirements of the filler metal specification ASME, sec.II, part C, AWS A5.28-05, classification ER80S-B6 when tested in accordance with that specification. Produced according to AWS A5.01, class S3

Ort / Town	Datum / Date		Abnahmebeauftragter / Authorized representative
Kapfenberg	02.01.2012	Dieses Zeugnis wurde maschinell erstellt und gilt ohne Unterschrift. This certificate was issued by DP-equipment and does not require signature.	Winkler